



Ephemeral Carbon

Carbon dioxide removal prepurchase application 2025

General Application

(The General Application applies to everyone; all applicants should complete this)

Public section

The content in this section (answers to questions 1(a) - (d)) will be made public on the [Frontier GitHub repository](#) after we announce purchases in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Company or organization name

Ephemeral Carbon

Company or organization location (we welcome applicants from anywhere in the world)

Santa Clara, CA, US

Name(s) of primary point(s) of contact for this application

Josh Perfetto, Bianca Broszus

Brief company or organization description <20 words

Ephemeral delivers scalable, affordable, and permanent carbon removal via rock weathering at sea, backed by a rigorous carbon quantification methodology.

1. Public summary of proposed project¹ to Frontier

- a. **Description of the CDR approach:** Describe how the proposed technology removes CO₂ from the atmosphere, including how the carbon is stored for > 1,000 years. Tell us why your system is best-in-class, and how you're differentiated from any other organization working on a similar approach. If your project addresses any of the priority innovation areas identified in the RFP, tell us how. Please include figures and system schematics and be specific, but concise. 1000-1500 words

Ephemeral is pioneering a novel, scalable, and ecologically responsible form of Ocean Alkalinity Enhancement called Oceanic Enhanced Weathering (OEW). We permanently remove atmospheric CO₂ by dispersing rapidly dissolving alkaline industrial byproducts across extensive continental shelf areas, combined with a rigorous and scalable Monitoring, Reporting, and Verification (MRV)

¹ We use "project" throughout this template, but the term is not intended to denote a single facility. The "project" being proposed to Frontier could include multiple facilities/locations or potentially all the CDR activities of your company.

framework that ensures accurate quantification of carbon sequestration.

OEW significantly improves upon conventional olivine-based Coastal Enhanced Weathering (CEW) by:

- **Accelerating Carbon Removal:** Sequestration occurs within years rather than decades.
- **Superior Scalability:** Leveraging expansive shallow continental shelves dramatically increases potential CO_2 removal capacity, overcoming the geographic limitations inherent to nearshore deployments.
- **Enhanced MRV Tractability:** Reduced sediment transport and calmer shelf waters facilitate clearer and more robust MRV methodologies

Overview

In OEW deployments, rapidly weathering alkaline materials are evenly dispersed across shallow continental shelves via self-unloading dry bulk cargo vessels. Natural ocean currents distribute these particles as they settle, and over the next 6–36 months, the particles react with seawater, consuming hydrogen ions and increasing alkalinity. This alkalinity converts dissolved CO_2 into stable bicarbonate and carbonate ions, creating localized CO_2 deficits. As alkalinity-enriched waters circulate and equilibrate with the atmosphere, additional atmospheric CO_2 is absorbed and permanently sequestered over geological timescales (exceeding 100,000 years). Targeted site selection, guided by advanced ocean modeling near upwelling regions or within currents favorable for rapid mixing, further enhances CO_2 absorption efficiency.

Ephemeral tailors each deployment to minimize ecological disruption and optimize CDR efficiency. Particle sizes are matched to native sediments, and the nutrient content is carefully managed to prevent unintended fertilization effects. Deployment conditions ensure optimal alkalinity release while preventing undesirable calcium carbonate or magnesium clay precipitation. Selecting deployment sites in warmer waters or regions with high remineralizing organic matter content further accelerates weathering.

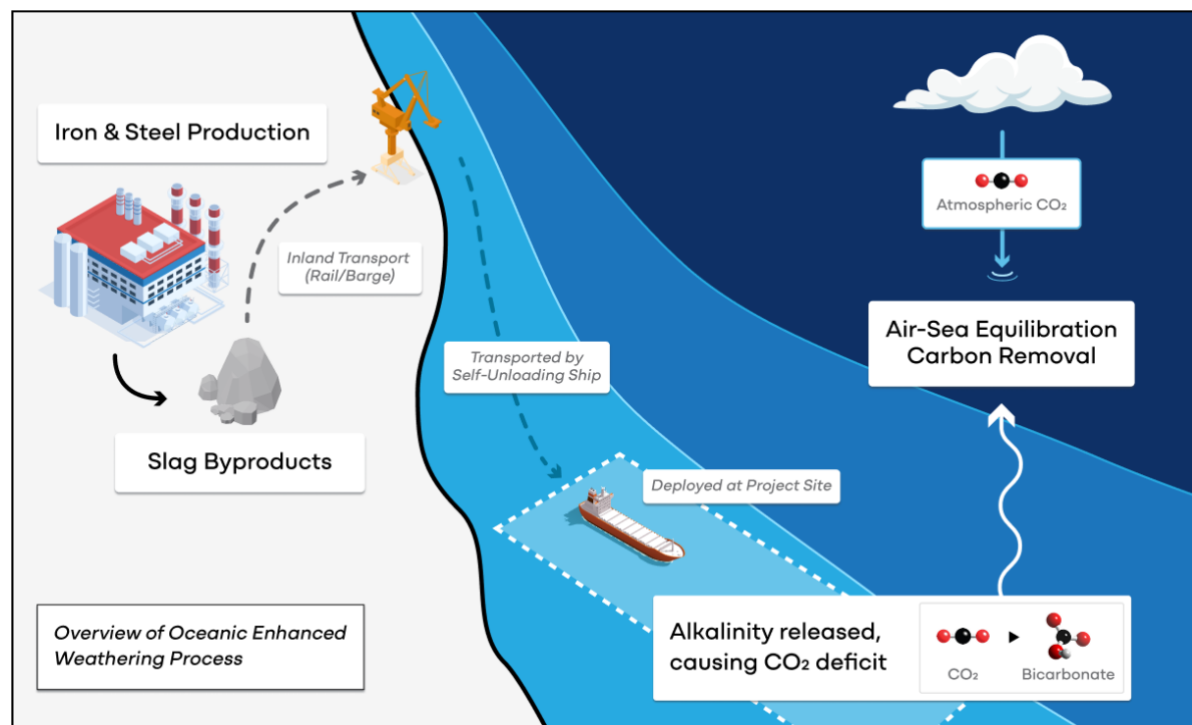


Figure 1: Overview of Oceanic Enhanced Weathering process

What Differentiates Us

1. Accelerated Carbon Removal Kinetics

Traditional Coastal Enhanced Weathering (CEW) typically employs sand-sized olivine, requiring decades for complete weathering and carbon sequestration, with even longer timelines in colder marine environments. Ephemeral differentiates itself by using rapidly dissolving industrial slags and finely ground silicate minerals. This innovation achieves significant carbon removal in less than three years, directly meeting urgent climate mitigation needs, credit delivery timeframes demanded by buyers, and enhancing financial viability.

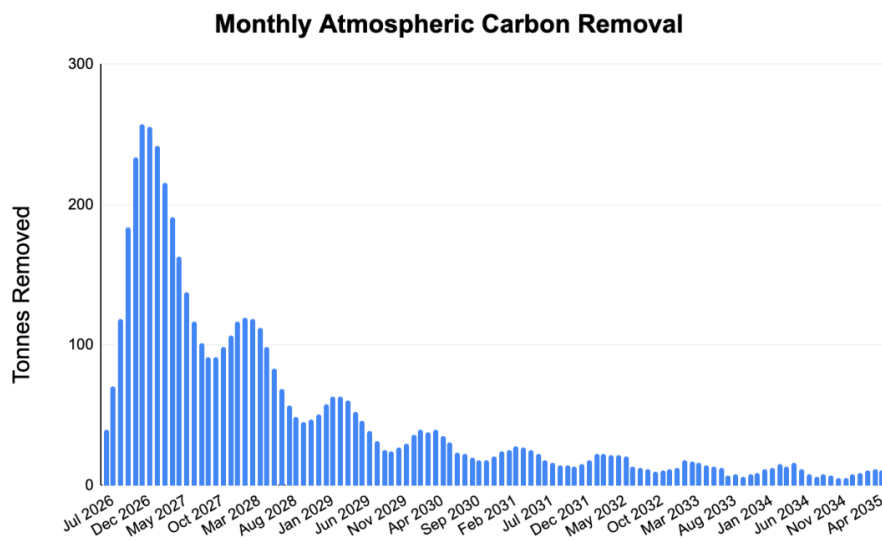


Figure 2: Anticipated monthly atmospheric carbon removal from pilot project

2. Robust, Cost-effective, and Scalable MRV Methodology

Weathering minerals is only half the equation; credible carbon removal verification and quantification is equally crucial. Measuring open-system carbon removal is particularly challenging at sea and over large deployment areas, where standard oceanographic methods (e.g., benthic flux chambers) would be impractical and cost-prohibitive.

Ephemeral's innovative MRV system solves these challenges by combining a reactive transport model with sediment-based monitoring:

- **Reactive Transport Model:** Simulates key sedimentary and mineral weathering processes (e.g., advection, diffusion, bio-irrigation, mineral dissolution, carbonate and Mg-silicate precipitation, sediment transport, TA/DIC fluxes to water column), while tracking natural soluble + insoluble tracers present in source materials.
- **Sediment Monitoring:** Sediment cores, easily retrieved from surface vessels, are analyzed by ICP-MS to quantify tracer concentrations and weathering signals. These data cost-effectively spatially scale monitoring and improve model accuracy by providing time-integrated measurements of the weathering process. Additional analysis is used in a confirmatory capacity.

Resulting alkalinity and DIC fluxes from the reactive transport model are then used to force ocean hydrodynamic and biogeochemical models, which model the ultimate air-sea equilibration, quantifying atmospheric carbon removal.

3. Enormous Deployment Capacity

Oceanic Enhanced Weathering (OEW) exploits vast shallow continental shelves—over 800 million hectares globally—alongside readily available alkalinity sources, surpassing Frontier’s >0.5 Gt CO₂/year criterion. Limiting factors are more likely tied to alkaline material availability rather than sea-floor area. Industrial slags not suitable for cement or other decarbonization uses offer the most accessible feedstock, with 350–450 million tonnes produced annually.

Global slag stockpiles—with over two billion tonnes in China alone—could further augment capacity, though older materials may contain higher trace metal concentrations hindering their use in OEW. For truly gigatonne-scale deployment, olivine-rich dunite reserves totaling hundreds of billions of tonnes provide long-term scalability. Other materials such as wollastonite, low-grade brucite, and additional industrial by-products could also be explored.

4. High Scalability with Minimal Infrastructure Requirements

Importantly OEW not only has a large ultimate potential, but is positioned to rapidly scale to meet that potential.

OEW using slags does not require land, significant energy inputs, or additional mining. By utilizing existing sand-grinding facilities and self-unloading dry bulk vessels, OEW can scale quickly to tens of megatonnes without major capital investments. Even at higher volumes, where investments in new ships and grinding infrastructure are required, capital intensity would remain quite low.

When finely ground olivine is used, specialized mineral-processing facilities with high-efficiency high mills become necessary, requiring some renewable energy but still far less capital- and energy-intensive than many other CDR solutions.

Altogether this is a technique that can grow rapidly in early years given the low capital and resource needs, and which can reach gigatonne scale in later years capital and resource-efficiently.

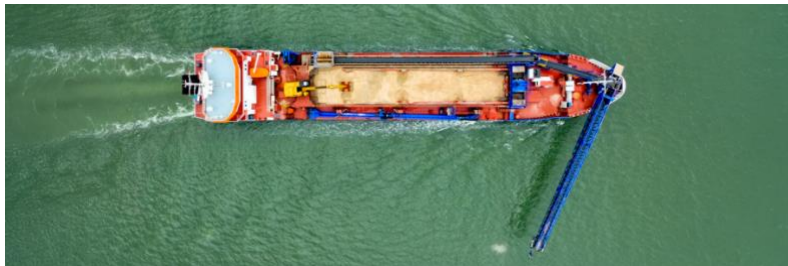


Figure 3: Selfunloading dry bulk carrier vessel unloading at sea

Alignment with Frontier mCDR Priority Innovation Areas

Our project directly addresses both Frontier priority innovation areas:

1. Scalable feedstocks for mineral OAE

Our initiative pioneers the use of industrial slags as feedstocks for marine Carbon Dioxide Removal (mCDR), representing a significant, readily available, and highly reactive alkaline resource. Prior to large-scale deployment, these materials require rigorous ecological validation. To ensure environmental safety and efficacy, our project incorporates comprehensive pre-deployment mesocosm experiments, alongside a robust ecological evaluation employing a Before-After-Control-Impact (BACI) study design. This methodology, incorporating a pre-deployment ecological baseline and two years of post-deployment monitoring, will assess impacts on critical marine communities, including phytoplankton, benthic infauna, and mobile benthic fauna.

2. mCDR approaches with clear co-benefits

Oceanic Enhanced Weathering (OEW), similar to other Ocean Alkalinity Enhancement techniques, holds promise for locally mitigating ocean acidification—potentially benefiting marine biodiversity such as fisheries and coral reefs. Additional anticipated benefits include enhanced silica availability, which could support silicifying organisms, improve fishery productivity, or even help mitigate harmful algal blooms (red tides).

Although these co-benefits remain theoretical and require validation through larger field trials, our pilot-scale deployment offers a unique opportunity to empirically investigate:

- **Plankton Community Responses:** Given the foundational role of phytoplankton and zooplankton in marine ecosystems, changes in their abundance, diversity, or productivity due to alkalinity enhancement could significantly influence local fisheries and broader ecological dynamics. Systematic monitoring will clarify these potential ecological impacts.
- **Localized De-acidification Impacts on Corals:** Observations of coral colonies within the deployment area and adjacent control sites will help determine if local alkalinity enhancements can positively influence coral health and resilience.

Though our initial deployment is intentionally limited in scale, the insights gained will yield critical ecological data, laying the groundwork for informed decision-making and responsible scaling of OEW technologies. This foundational work will ultimately enable the larger deployments necessary to validate and realize co-benefits at fishery and regional scales.

- b. **Project objectives:** What are you trying to build? Discuss location(s) and scale. What is the current cost breakdown, and what needs to happen for your CDR solution to approach Frontier's cost and scale criteria?² What is your approach to quantifying the carbon removed? Please include figures and system schematics and be specific, but concise. 1000-1500 words

Ephemeral Carbon is developing the world's first scalable, cost-effective, and community-driven Oceanic Enhanced Weathering (OEW) project. By accelerating the natural process of rock weathering in the ocean, we permanently remove atmospheric CO₂ through the dispersal of rapidly dissolving alkaline industrial byproducts across shallow coastal waters.

Our system couples existing industrial waste streams (iron and steel slags) with open-water vessel deployment and a novel, scalable MRV framework to achieve:

- *Fast carbon removal:* Sequestration within 6–36 months instead of decades.
- *Low cost:* A clear pathway to <\$100/ton CO₂ at scale.
- *Scalability:* Gigatonne/year potential leveraging vast continental shelf areas and abundant industrial byproducts around the world.
- *Community and ecosystem alignment:* Significant ecological co-benefits (such as localized ocean deacidification) at scale and permanent investment in local capacity
- *Scientific credibility:* Data sharing, open science, and rigorous field validations.

We aim to accelerate climate impact by combining cutting-edge marine science, community partnerships, and strategic use of existing industrial infrastructure to rapidly scale Oceanic Enhanced Weathering as a trusted carbon dioxide removal solution.

² We're looking for approaches that can reach climate-relevant scale (about 0.5 Gt CDR/year at \$100/ton). We will consider approaches that don't quite meet this bar if they perform well against our other criteria, can enable the removal of hundreds of millions of tons, are otherwise compelling enough to be part of the global portfolio of climate solutions.

Location and Scale

We are prioritizing deployments in communities that are disproportionately affected by climate change, where coastal livelihoods are increasingly vulnerable. Our first pilot project will take place at a site in the Caribbean, leveraging the region's extensive shallow coastal areas, favorable ocean conditions, and strong climate resilience priorities. This project is designed to validate our Oceanic Enhanced Weathering (OEW) methodology, test our MRV framework, and assess ecological safety through controlled, small-scale deployment.

The pilot will involve dispersing 10,000 tons of alkaline feedstock - specifically, industrial slags, a byproduct from the iron and steelmaking industry - across a selected contiguous site along the continental shelf. This initial scale is sufficient to validate our deployment configuration and MRV framework, while remaining small enough to thoroughly monitor ecological impacts and build regulatory and community trust.

Community engagement is a central component of our approach. We are committed to ensuring that our work delivers meaningful, lasting benefits to the coastal communities where we operate. In addition to helping advance national and regional climate goals, we aim to build local capacity through research partnerships, internship-to-career pathways, and knowledge sharing. Through this project we hope to provide a positive example of how carbon removal projects can support local priorities alongside global climate efforts.

Following the pilot, we plan to expand to additional sites across the Caribbean and beyond. Each subsequent deployment will incrementally increase the quantity of feedstock and the spatial scale of deployment, supported by lessons learned from earlier projects. Our long-term vision is to achieve gigatonne annual CO₂ removal by operating multiple regional hubs near slag/mineral sources and coastal shelf sites, allowing Oceanic Enhanced Weathering to scale safely, efficiently, and globally.

Current Project

While extensive laboratory studies and mesocosm experiments have already shown the promise of using iron and steel slags for oceanic carbon removal, no simulation can fully capture the complexity of real-world ocean conditions. As emphasized by Frauke Kracke at the March 2025 Ocean Visions Biennial Summit in Vancouver, "We need more OAE deployments. We need success stories." We must demonstrate - through actual, monitored deployments - that Oceanic Enhanced Weathering (OEW) can be performed cost-effectively, safely, and at scale.

Our first project's cost structure reflects the realities of early-stage deployment: high upfront expenses that are not yet amortized across multiple projects, along with conservative assumptions for monitoring, reporting, and verification (MRV) activities.

A significant portion of the cost is dedicated to building and validating our MRV framework, which requires intensive monitoring, modeling, and third-party verification to ensure scientific credibility. This MRV burden is expected to drop substantially as we gain field validation, refine our models, and establish standardized protocols.

Another notable cost component is the transportation and logistics of bringing slag feedstock from production sites to deployment locations. As we grow, we plan to optimize supply chains by sourcing materials closer to project sites and co-locating processing hubs with slag production centers, reducing both financial costs and lifecycle emissions.

Overall, the first deployment is designed not only to achieve carbon removal, but also to de-risk and de-bottleneck the pathway to future projects, where economies of scale, streamlined MRV, and supply chain optimization are expected to drive costs down significantly. This will provide a clear path toward <\$100/ton CO₂ removal.

Projects at Scale

Three major steps are needed for us to reach Frontier's cost (<\$100/ton) and scale (>0.5 Gt/year) criteria:

1. Validation and Streamlining of MRV:

Our early projects require intensive and conservative MRV to establish scientific rigor and credibility. As we validate our models and field methodologies through pilots, we will significantly streamline MRV, moving from highly customized, resource-intensive frameworks to standardized, efficient protocols. This will sharply lower per-ton monitoring costs, one of the largest initial cost drivers.

2. Transportation and Supply Chain Optimization:

At pilot scale, material transportation and deployment logistics are relatively inefficient. Scaling will involve sourcing slag feedstocks closer to deployment ports, establishing local processing hubs, and using larger vessels for ocean transport and deployment. This will reduce inland transport, grinding, and shipping costs by leveraging economies of scale and better geographic integration between slag sources and deployment sites.

3. Larger, Integrated Deployment Projects

Our early projects are small by design to focus on safety and validation. Future deployments will involve larger volumes of feedstock dispersed over wider areas, allowing for the amortization of equipment costs across far greater tons of CO₂ removed. At a larger scale, the cost impact of one-time permitting, and initial environmental impact assessments will also diminish per ton.

It is important to note that no fundamental technology breakthroughs are required for successful execution of Oceanic Enhanced Weathering at scale. The vessels, grinding equipment, and feedstock materials all exist today.

What is needed is execution - demonstrating ecological safety, social acceptance, and validation of open ocean MRV robustness - to unlock the pathway to low-cost, gigatonne-scale carbon removal.

Monitoring, Reporting, and Verification (MRV)

Measuring open-system carbon removal is particularly challenging at sea and over large deployment areas, where standard oceanographic methods (e.g., benthic flux chambers) would be impractical and cost-prohibitive. Additionally, since Oceanic Enhanced Weathering (OEW) creates subtle changes against a highly variable ocean background, direct measurement alone is insufficient.

Our approach combines controlled field releases with a multi-layered modeling and measurement framework, validated through laboratory experiments and field observations, to rigorously track and verify the fate of added alkalinity and the resulting CO₂ removal:

- *Material Release and Dissolution Characterization:* We precisely track the amount of alkaline material deployed and determine its dissolution behavior through laboratory experiments using natural seawater. This ensures accurate inputs for subsequent modeling.
- *Near-Field Modeling and Reactive Transport Simulation:* We use reactive transport models to simulate key sedimentary and mineral weathering processes, including advection, irrigation, dissolution, and precipitation. These models track the fate of tracers embedded within the deployed material to validate dissolution and dispersal dynamics.
- *Field Monitoring and Sediment Sampling:* Sediment cores, easily retrieved from surface vessels, are analyzed via ICP-MS (Inductively Coupled Plasma Mass Spectrometry) to measure tracer concentrations and weathering signals. These time-integrated data points improve the calibration and accuracy of our reactive transport models.

- *Oceanic Carbon Flux Modeling:* The resulting outputs - alkalinity and dissolved inorganic carbon (DIC) fluxes - are then used to drive larger-scale hydrodynamic and biogeochemical ocean models. These models simulate air-sea CO₂ exchange, quantifying the amount of atmospheric CO₂ absorbed as a result of the created CO₂ deficit.
- *Final Quantification of Carbon Removal:* By linking material deployment, dissolution, tracer tracking, and air-sea flux modeling, we create a fully traceable, scientifically robust chain of evidence for permanent atmospheric carbon removal.

Overall, our MRV approach is designed to balance scientific rigor with practical scalability, enabling credible, third-party-verifiable carbon removal at oceanic scales.

- c. **Risks:** What are the biggest risks and how will you mitigate those? Include technical, project execution, measurement, reporting and verification (MRV), ecosystem, financial, and any other risks. 500-1000 words

1. Obtaining community support

Trust is critical in coastal communities that have historically been burned by large international development projects that promised community co-benefits but failed to deliver. As a result, there is natural and understandable skepticism toward projects perceived as extractive or environmentally harmful. The biggest risks in obtaining community support for our Oceanic Enhanced Weathering (OEW) project include a lack of understanding or trust in ocean-based carbon dioxide removal (CDR) methods, concerns about ecosystem health and coastal livelihoods, and perceptions of top-down decision-making without sufficient local involvement.

Mitigation: We are implementing a community-first approach that integrates public engagement, transparency, and communication of co-benefits from the outset. Early and continuous engagement with coastal stakeholders, including fishers, tourism operators, and local NGOs, ensures we listen and respond to concerns, share data openly, and shape the project around community needs and values. We are partnering with local research institutions and government agencies to build credibility and promote two-way knowledge exchange. We are also hiring local project leadership to spearhead our efforts and establishing a permanent presence in the countries where we operate, signaling our long-term commitment to collaboration and community investment.

Our project siting efforts across multiple Caribbean countries have already received an overwhelmingly positive response from stakeholders, with local leaders expressing strong interest and offering guidance on the various stakeholder groups that should be consulted. Communities in the Caribbean, which have been disproportionately impacted by climate change, are eager to explore solutions that can strengthen their marine ecosystems, particularly given the critical role of fisheries and coastal resources in local livelihoods. We will provide regular updates on project activities, open-access data, and marine ecosystem safeguards through community briefings, Q&A sessions, and online forums to maintain transparency and trust throughout the project lifecycle. Additionally, we are working in alignment with national climate action plans, where priorities such as supporting biodiversity, expanding ocean research, and advancing climate resilience solutions are already well established. We will also clearly communicate how the project will generate tangible local benefits, including job creation, internship-to-career pathways, eco-tourism opportunities, and fisheries improvements.

2. MRV certainty

We have developed our MRV methodology based on published literature, existing data sets, and our own laboratory experiments. While we anticipate that the field pilot will validate this approach, there remains a risk that the MRV methodology may be more challenging to validate or yield higher quantification uncertainty than expected, potentially reducing the number of deliverable carbon credits.

Mitigation: To address this risk, we have built a substantial 12-month buffer into our timeline between initial data availability and our committed credit delivery date to Frontier. This period allows ample time for iterative refinements of methodologies and models once empirical data become available. We also have developed an alternative, simpler methodology (though expected to have higher long-term costs/uncertainty) which could be employed if our preferred approach proves to require more time to validate.

3. Ecological safety

There is a risk that our pilot deployment could result in negative ecological impacts, which could impede our ability to scale future projects.

Mitigation: To proactively minimize ecological risks, we are implementing thorough pre-deployment mesocosm testing designed to establish safe operational parameters. Our pilot design includes multiple adjustable parameters—such as particle size distribution and deployment thickness—that can be modified to ensure ecological safety. Furthermore, we have adopted a conservative approach by selecting the least risky slag feedstock available and intentionally avoiding aggressive optimization of CDR rates. These measures collectively minimize the likelihood of significant ecological impacts from

our pilot.

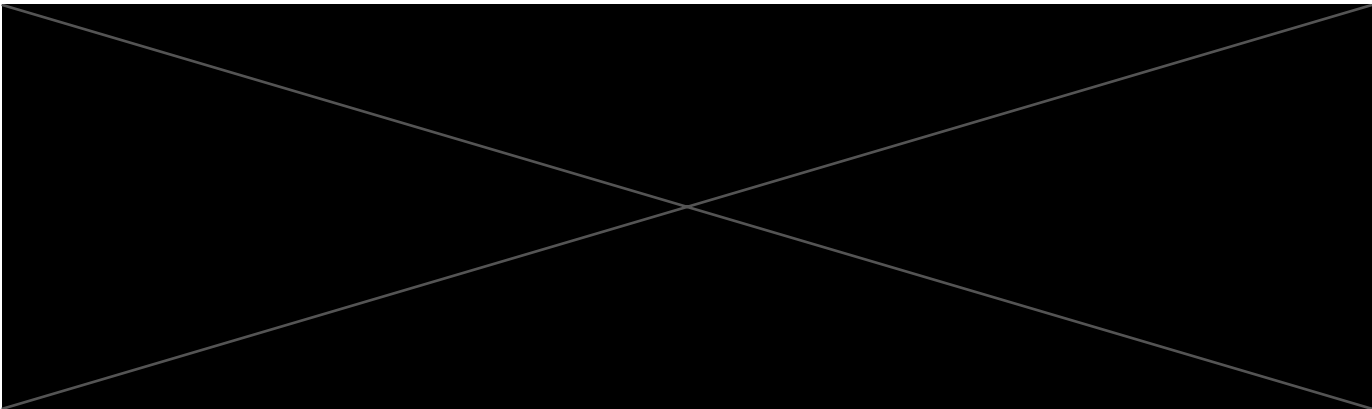
4. Project financing

Given the intensive MRV efforts required for the initial pilot to validate both methodology and ecological safety, MRV-related costs will initially be high. A significant risk is that these costs may not decline rapidly enough to ensure the financial viability of future projects under prevailing voluntary market conditions.

Mitigation: To manage this financial risk, we prioritize demonstrating the speed, scalability, and quantifiability of our approach, aiming to attract venture capital investment to support subsequent projects and reduce MRV costs. Additionally, we are actively seeking multiple sources of non-dilutive funding as described in section 2(k). We are also initiating targeted sales and marketing efforts to engage aligned brands and small to medium-sized businesses (SMBs), thereby expanding both the voluntary carbon market and our customer base. Advancing our technology readiness level (TRL) remains a core focus, as doing so will enable us to secure offtake agreements, thereby facilitating potential debt financing opportunities.

- d. **Proposed offer to Frontier:** Please list proposed CDR volume, delivery timeline and price below. If you are selected for a Frontier prepurchase, this table will form the basis of contract discussions.

Proposed CDR over the project lifetime (tons) <i>(should be net volume after taking into account the uncertainty discount proposed in 5c)</i>	310
Delivery window <i>(at what point should Frontier consider your contract complete? Should match 2f)</i>	July 2027
Levelized cost (\$/ton CO ₂) <i>(This is the cost per ton for the project tonnage described above, and should match 6d)</i>	\$1,347
Levelized price (\$/ton CO ₂) ³ <i>(This is the price per ton of your offer to us for the tonnage described above)</i>	\$1,613



³ This does not need to exactly match the cost calculated for “This Project” in the TEA spreadsheet (e.g., it’s expected to include a margin and reflect reductions from co-product revenue if applicable).